Lab 1 Structural Mechanics

ASEN 3802: Aerospace Sciences Lab 2
University of Colorado at Boulder

Please refer to Canvas for lab groups and supporting documents.

Assigned Date: Week 1, 2026
Due Date Part 1: Wednesday Week 4 January 28th, 2026
Due Date Part 2: Wednesday Week 5 February 4th, 2026
Due Date Part 3: Wednesday Week 6 February 11th, 2026

Learning Outcomes and Policies

Learning Outcomes:

1. Understand the derivation of a equivalent beam model and the uses for equivalent models.
2. Introduce finite element analysis and its applications.
3. Quantify the impact of assumptions on modeling compared to experimental results.
4. Compare the two modeling methods to experimental data.

Group Collaboration Policy:

For this laboratory assignment students will be organized in nominal groups of three or four to complete all of the problems or questions posed below. Interactive collaboration within the group is expected on each of the laboratory assignment components instead of individual division of the effort so that every student has a complete and thorough understanding of all elements of the laboratory assignment. All group members are responsible for timely communication and organization of the work to be done. In the event a member is not communicating, the team should let the instructor know and discuss options.

Outside of the group you may discuss the means and methods for formulating and solving problems and even compare answers with other groups, but you are not free to copy work from another group or from prior semesters. *Copying material, including code and algorithms, from any resource (including solutions manuals) and submitting it as one's own is considered plagiarism and is an Honor Code violation. If students are found in violation of plagiarism of any amount they will receive a zero on the entire assignment and will be written up for an Honor Code Violation for further sanctions.*

Groups are encouraged to set-up a shared folder accessible to the entire group, but no one outside the group. By regularly communicating and sharing progress among the entire group all students should be empowered to fully understand every element of the laboratory assignment and will be equally responsible for completing and submitting the assignment in a timely manner. Furthermore, sharing the up-to-date progress between all group members will ensure redundancy in case extenuating circumstances arise (i.e. a group member is sick or otherwise unavailable during a lab period or on the assignment deadline).

Individual Contribution Evaluation:

All team members will rate the performance of their partners with a numerical value, as well as comments explaining deductions. This will be collected anonymously. The performance of each group member is rated with a "contribution factor" on a scale of 0 to 100%. A score of 100% indicates that the member contributed the expected share to the computational assignment (code development and data analysis) and to the preparation of the plots and discussions.

Note that this is not done for each assignment, but for each **lab**. It is not expected that each member contribute exactly equally for each assignment, but for the overall lab.

Assignment Overview:

Project Description

In this lab you will analyze the static load test of a 16-bay space (3D) truss in a beam-like support-loading configuration. Experimental measurements are to be compared with analytical modeling, as well as simulation results obtained using a general-purpose, commercial Finite Element Method package (ANSYS). The FEM model is used to quantify the impact of modeling assumptions and uncertainties in geometry, material properties, and measured data. This document provides general information for this lab. It does not include details on the experimental procedure or ANSYS tutorials, which are provided separately. The lab is divided into three parts, each with their own submission to keep you on track during the semester.

ANSYS is installed on all computers in the PILOT and co-PILOT. There is also a free student version of ANSYS that can be downloaded on personal computers and used for this lab; see https://www.ansys.com/academic/free-student-products. Remote to access ANSYS is available at https://www.colorado.edu/engineering/cloudcomputing. The ANSYS tutorial provides more information on remote access.

Submission Requirements

There is no formal lab report required for this lab. Instead, the lab is broken up into three parts that will be graded separately. For part 1 the analytical beam bending solution for the three loading cases will be developed. For part 2 the experimental results will be plotted and the analytical solution will be used to determine the location of the loads on the truss. In part 3 the truss will be modeled with FEM in ANSYS and compared to the experimental results.

There is no specific format required for the submissions. Simply submit your **Derivations, Plots/Tables, and Discussions** as denoted in each part. Note that while there is no format required, you will be graded on the construction of your plots as well as the clarity of your discussion. Regarding presentation, make sure that you follow these guidelines:

- All plots should be readable. This includes using different color or line styles and a suitable font size for the axis labels and all other text in the plot. The range of both axes should be chosen to focus on the region of interest (i.e., the data).
- Show your work, including equations used and partial results.
- All results should be presented with appropriate units.
- Be quantitative when comparing results. Use percentage of error or deviation.

Grading Breakdown

Part	Due	Description	Item	Weight
Part 1	Wednesday Week 4	Task 1	Moment of Inertia Calculation	5%
		Task 2	Centered load Equivalent Beam Model	5%
		Task 3	Off-centered load Equivalent Beam Model	5%
		Task 4	Two Load Equivalent Beam Model	5%
Part 2	Wednesday Week 5	Task 1	Experimental Data Analysis	15%
		Task 2	Determine Location of Off-centered Load Case	10%
		Task 3	Determine Location of 2 Load Case	5%
Part 3	Wednesday Week 6	Task 1	FEM Modeling Results	10%
		Task 2	Comparison and Discussion of the Three Analysis Methods	20%
		Task 3	Uncertainty Quantification	20%
Total				100%

Experiment and Hardware

Test Article

A photo of the 16-bay truss structure is shown in Figure 1. SI units are recommended, although some dimensions and forces are initially expressed in English units. The truss consists of 16 cubic bays with joint-to-joint distances of 0.250 m (250 mm). The struts are fabricated from extruded 6061-T6 Aluminum tube stock as illustrated in Figure 2. The cross section dimensions are $d = 3/8$ in $= 9.525$ mm and $t = 1/16$ in $= 1.587$ mm. The elastic modulus for 6061-T6 Aluminum typically ranges between 69 and 70 GPa.

Applied loads (for the reference configuration) and supports are illustrated in Figure 4. The button load cells used to measure the reaction loads at either end of the truss allow truss ends to rotate and therefore provide nominally pinned and roller boundary conditions. For the main configuration, a total load of $P = 50$ lbs $= 222.4$ N will be applied with shot bags at the midspan of the truss. This can be modeled as two point loads of $P/2$ applied on the two top joints at midspan. **Two other cases will be considered. First, the same load, but applied not at the center. Second, a loading case in which two different nodes are loaded symmetrically.**

The internal force across one of the longeron struts on the lower side of the truss near the midspan will be measured by an inline load cell. Vertical displacements will be also measured at the midspan using a Linear Variable Differential Transformer (LVDT).

Note on Truss Fabrication

The joint-to-joint dimensions are **metric** because the original trusses, fabricated in 1992, were supposed to be a scaled model of the main truss components of the original International Space Station (ISS) design of 1991. This design was later modified. To reduce costs, however, struts were fabricated from commercial aluminum stock, cut and precisely machined at both ends to fit into the joint attachments. Ball joints and strut fittings were custom made at the Physics shop.

Experimental Measurements

The structure will be loaded using shot-put bags of 10 lbs, in 10 lbs increments up to the maximum total load. The truss is placed on load cell support plates at both ends. Several sets of measurements will be taken. The midspan deflection

Fig. 1 Photograph of the 16-bay truss test article.

of the truss will be recorded with an LVDT device. Reaction forces will be recorded through the load cells under the support plates. The internal force in a longeron member adjacent to midspan will be measured through an in-line load cell. Please be sure to convert all units to the same system before calculations and comparisons.

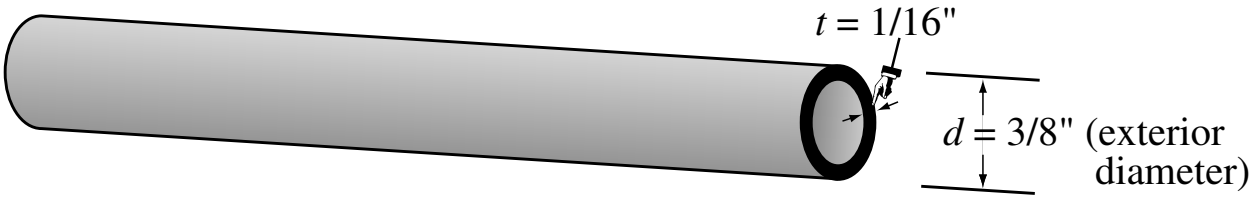

Fig. 2 Truss member (strut) cross section schematics. End fitting details (threads and joint adapters) not shown.

Part 1: Equivalent Beam Calculation

You will first model the truss using an *equivalent beam* model. This model treats the truss as beam where the moment of inertia is computed via parallel axis theorem considering only the four longeron-strut cross sections taken with respect to the neutral axis of the equivalent beam. The moment of inertia of the individual struts about their individual bending axes, i.e. the member self-inertia, is negligible. This highly simplified model ignores the presence of diagonal members. The elastic modulus is that of 6061-T6 aluminum alloy.

Part 1 Task 1: Calculate Moment of Inertia of the Equivalent Beam

5% of overall grade - DERIVATION/CALCULATION
Derive an expression for the area moment of inertia of the truss cross section. Present the equations used and final numeric answer.

Part 1 Task 2: Equivalent Beam Model with Centered Load

5% of overall grade - DERIVATION/CALCULATION
Derive an expression for the mid-span deflection and expected stress at the inline load cell when a load is placed in the center of the equivalent beam.

Part 1 Task 3: Equivalent Beam Model with Off-centered Load

5% of overall grade - DERIVATION/CALCULATION
Derive an expression for the mid-span deflection and expected stress at the inline load cell when a load is placed at an arbitrary position along the equivalent beam.

Part 1 Task 4: Equivalent Beam Model with 2 Symmetric Loads

5% of overall grade - DERIVATION/CALCULATION
Derive an expression for the mid-span deflection and expected stress at the inline load cell when two equal loads are placed symmetrically on either side of the equivalent beam.

Part 2: Experimental Results

Part 2 Task 1: Analysis of Experimental Data

15% of overall grade - 10% PLOTS, 5% Discussion

PLOT and DISCUSSION

Provide plots of the reaction forces, displacements, and internal force measurements, for all cases presented. Plot each type of measurement for all cases in the same plot, in all cases as a function of the applied loading (i.e., plot the linear displacement for all cases in one plot).
Compare the behavior of each loading case, and discuss the results. Do they make sense?
Use linear regression and uncertainty analysis to verify the linearity of the measurements with respect to the external load magnitude. Explain the method of linear regression used and provide a plot of the regression curve. How do you confirm that the regression is linear?

Part 2 Task 2: Determine location of off-centered load case

10% of overall grade - 5% CALCULATION, 5% DISCUSSION
Consider second the case in which the load is applied at a single node, not in the center.

CALCULATION and DISCUSSION

Can you identify the position of the loading, based on the reaction forces? Why? If not, use other information as needed, and explain the process. Show the position of the load. Compare the displacement and internal force from the experimental results and with the results from the equivalent beam model.

Part 2 Task 3: Determine the location of the symmetric two load case

5% of overall grade - 3% CALCULATION, 2% DISCUSSION
Consider now the third case, in which the load is applied at two different nodes symmetrically, with the same load at each of them.

CALCULATION and DISCUSSION

Can you identify the positions of the loads based on reaction forces? Why? If not, use other information as needed, and explain the process. Show the position of the loads. Compare the displacement and internal force from the experimental results and with the results from the equivalent beam model.

Part 3: FEM Results and Comparison Between Methods

For this part, the reference loading case with the load centered on the truss will be modeled with FEM and compared to the experiment and equivalent beam model from Parts 1 & 2.

The FEM Structural Model

Figure 3(a-b) shows an schematic of how the Finite Element software will discretize the structure as a series of nodes and bars (node numbering will likely be different). A typical (repeating) two-bay unit is illustrated in Figure 3(c) — the full model is generated by repeating this unit seven times. You will need to use self-consistent units for the length, loads, and stress/elastic moduli. For the purposes of this lab we will not explicitly model the details of the mechanics of the node assemblies, for example threads and joint fittings. This potential source of modeling error, however, should be considered when comparing experimental and computational results and performing uncertainty analysis. More details on how to define the model will be presented in the ANSYS tutorial.

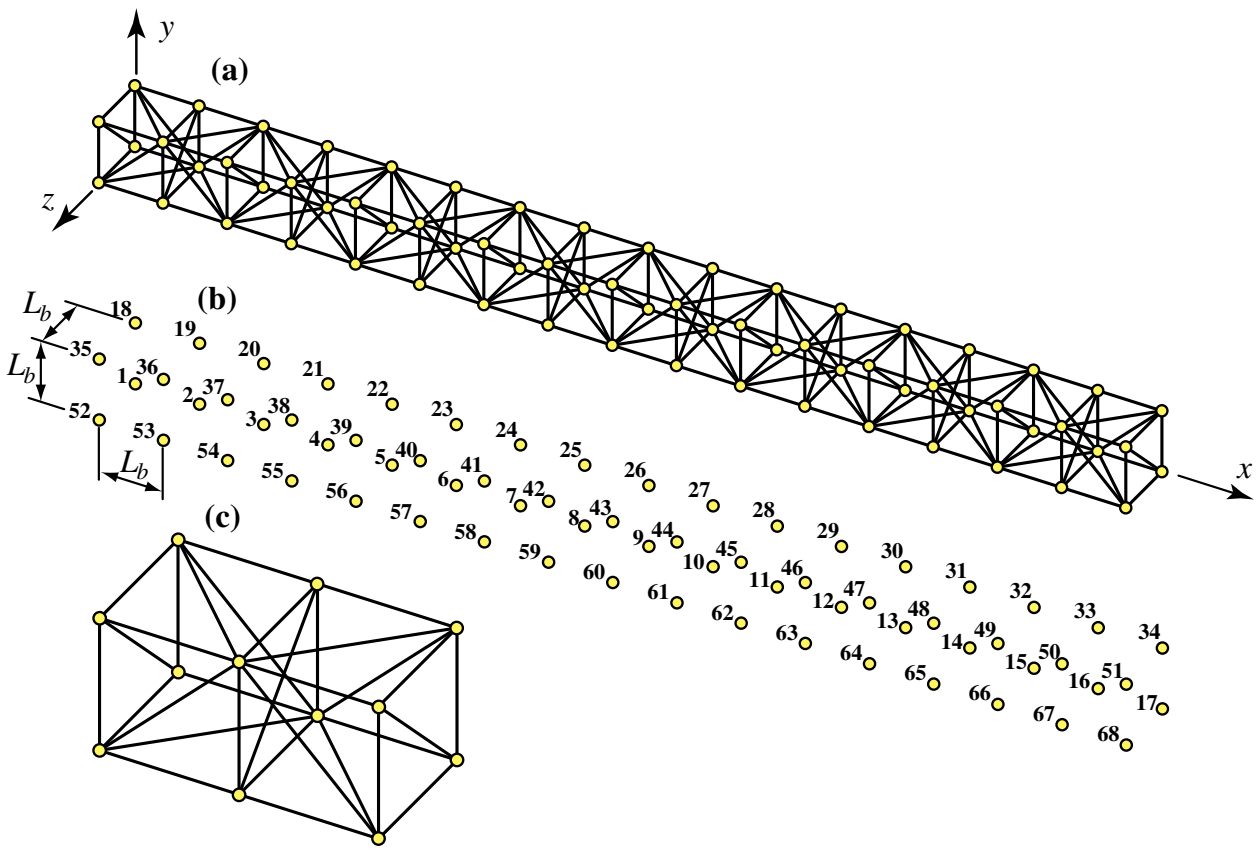

Fig. 3 Schematics of the 16-bay space truss FEM model: (a) 16-bay truss showing members and nodes; (b) node numbering detail (the numbering for the ANSYS model will likely be different); (c) repeating 2-bay unit used to generate the full truss.

The applied loads and support conditions are schematically shown in Figure 4. In the reference case, the maximum load to be applied at midspan is P = 50 lbs = 222.4 N. The load will need to be approximated in the model as load on the nodes. It is recommended to fix the left-support nodes in all three (x, y, z) directions. The right-support nodes are fixed in the transverse (y, z) directions but allowed to move in the longitudinal (x) direction, as illustrated in Figure 4. This modeling decision approximates a friction-less support and tries to avoid buildup of spurious axial forces in the FEM model. However, friction may cause real axial forces. Thus, the influence of assumptions for modeling the boundary conditions should be studied in the uncertainty analysis.

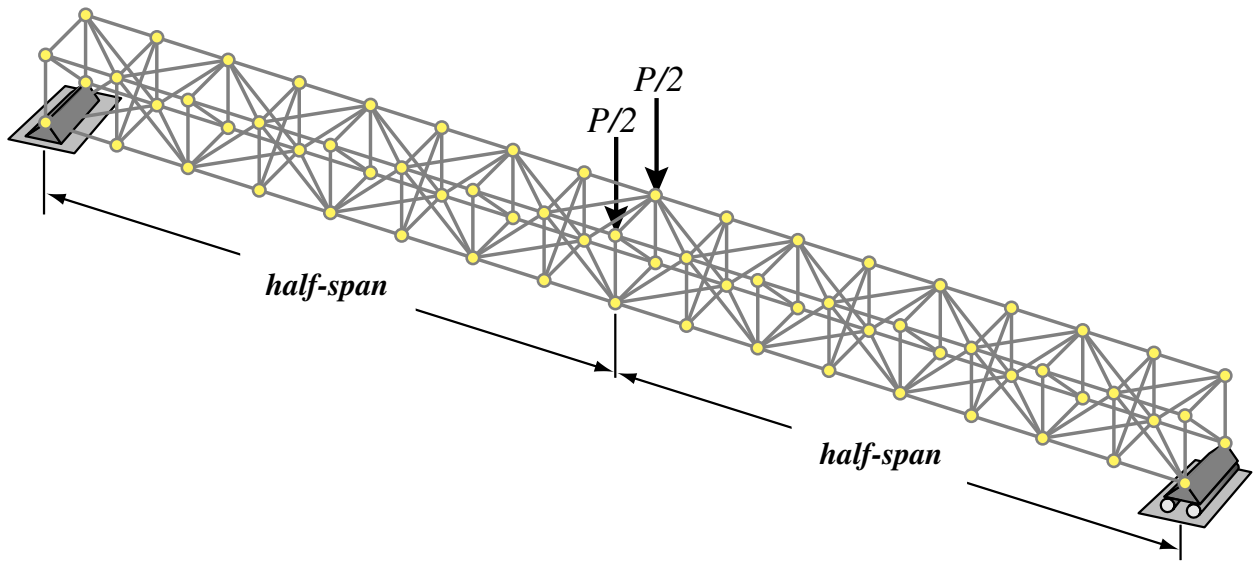

Fig. 4 Applied loads and support conditions. These are idealizations of the real loads and supports: the shot-put-applied loads actually span two bays, whereas the fixed and rolling simple-support conditions idealize the actual supports.

In the simulations you should record the set of node displacements and internal (axial) member forces computed by ANSYS under a load of P = 50 lbs = 222.4 N. From this set of results, you can select appropriate quantities to compare with measurements. Note: To compare predictions with a different load P, a simple scaling is sufficient because the FEM truss behavior is assumed to be exactly linear: if the load is doubled, displacements and internal forces are doubled.

Part 3 Task 1: Plot Results from FEM

10% of overall grade - 7% PLOT, 3% DISCUSSION

PLOT and DISCUSSION

Using the finite element model, plot the undeformed versus deformed truss. Provide a plot of the internal forces in the bars, and discuss if the trend is to be expected.

Part 3 Task 2: Comparison with analytical and FEM results

20% of overall grade - 10% TABLE, 10% DISCUSSION

TABLE and DISCUSSION

Compare the experimental results in the reference configuration (load in the center) with the predictions obtained with the finite element and the equivalent beam model. Tabulate the reaction forces, displacement, and internal forces to compare to the experimental results. Provide *quantitative* measures of the differences. Discuss the reasons for the discrepancies, if you observe any.

Part 3 Task 3: Uncertainty analysis

20% of overall grade - 10% TABLE, 10% DISCUSSION

TABLE and DISCUSSION

Finally, explore potential sources that may cause the differences between experimental measurements and FEM results. To provide a *quantitative* assessment of these differences, analyze the influence of the following uncertainties by performing additional FEM simulations. Present your results in form of tables and/or graphs. You need to address all of the following possible sources of error:

- Imperfect joints, such as free-play, may reduce the effective stiffness of the bars.
- Free-play in the in-line load cell may also reduce the effective stiffness of the bar containing the in-line load cell.
- Friction at supported joints may require considering different boundary conditions.
- Manufacturing imperfections may change the relative position of the joints and the cross-section of the struts.
- The bar model assumes a uniform axial stress distribution in the struts; thus, the struts have no internal bending moments and the joints are modeled as perfect pin-joints. Consider that the joints can transmit bending moments and the struts provide bending stiffness, i.e., act as beams.

Be as quantitative as possible when characterizing the effect of each source of uncertainty. Discuss their relative importance.

Grading Rubrics:

DERIVATION/CALCULATION Grading Rubric

Item	Weight	Grading Criteria
Accuracy	60%	Is the algebraic expression correct? If applicable, is the numeric value correct and include proper units?
Derivation/ Explanation	30%	Are key steps shown in the derivation? (Note all steps are not required) Is it clear to the reader the method utilized to reach the final expression?
Format	10%	Are final results typeset? Are derivations legible and easy to read? (handwritten is okay)
Total	**100%**	

PLOT Grading Rubric

Item	Grading Criteria
Plot Data - 60%	
Plot Data	Is all expected data included? Is the data that is included correct? Is there any data included that is not relevant?
Plot Format - 40%	
Axes	Are the axes properly labeled (with units)? Are the axes properly scaled and divided?
Linetypes and Markers	Do the color choices of the lines add/detract from the clarity of the plot? Are the markers appropriate for the data plotted? Do the line weights and format add/detract from the clarity of the plot?
Caption	Does the caption just repeat the axis information? Does the caption give all necessary information for the plot to stand alone? Is the caption well-formatted and concise?
Annotations and Legend	Is the legend properly formatted and limits set? Do annotations (if present) improve/detract from the plot? Is the plot over/under annotated?
Overall Format	Does the font size of the plot match the rest of the paper? Is the plot legible and clear with no obscuration of data? Is the plot suitable for publication in a professional setting?
Total	**100%**

DISCUSSION Grading Rubric

Item	Weight	Grading Criteria
Content	60%	Have all the questions asked been addressed? Are the answers to the questions fundamentally correct and complete? Does the discussion show a deeper understanding of the material?
Structure	20%	Is the discussion concise and limited to 1-2 paragraphs? Do the paragraphs have a proper structure?
Writing	20%	Is the text free of grammar mistakes and typos? Has the writing been edited for clarity, brevity, word choice, and conciseness?
Total	**100%**	

Contribution Report - Structural Mechanics

Lab member 1:

Summary of work completed

Lab member 2:

Summary of work completed

Lab member 3:

Summary of work completed

Lab member 4:

Summary of work completed

Lab member 5:

Summary of work completed